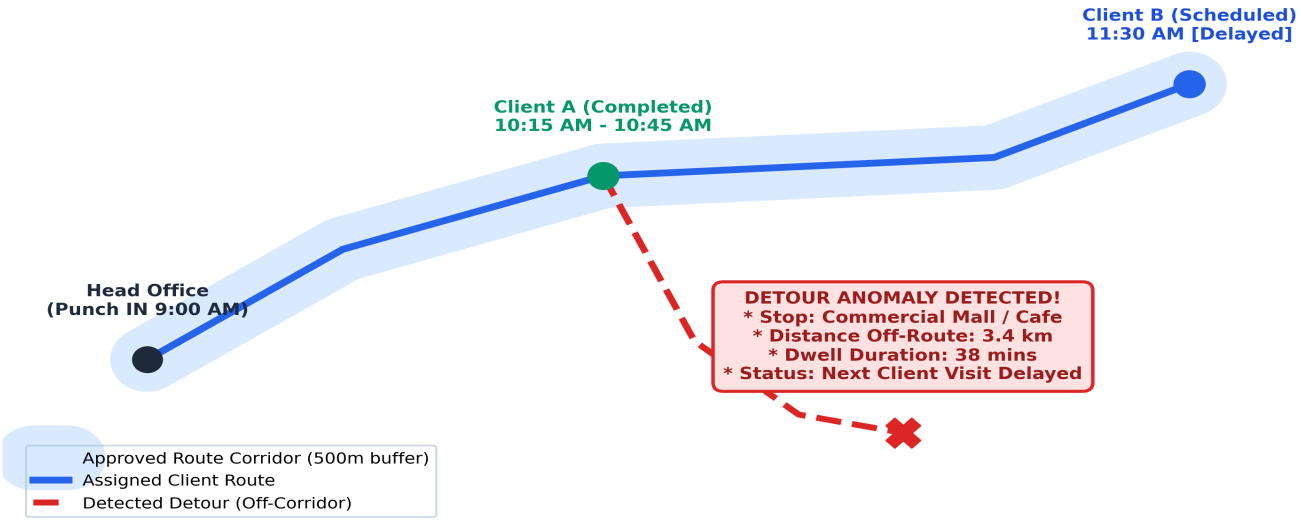
PROJECT: AI Workforce Anomaly Tracking	**VERSION:** 1.0 (Executive Pitch)	**DATE:** October 2026	**STATUS:** Proposed Architecture

AI-Powered Field Worker Detour & Anomaly Monitoring

Automated Route Deviation Detection, Personal Stop Tracking & Real-Time Admin Alerts

Executive Answer to Client: YES, this is fully possible and highly practical. By fusing mobile GPS breadcrumbs with an intelligent Geo-Corridor algorithm and Google Gemini AI, the system continuously monitors field workers during active duty hours. When a worker makes an unauthorized deviation from their scheduled customer path or halts at an unassigned location for personal activities, the AI instantly validates the context and dispatches a detailed alert to company management.

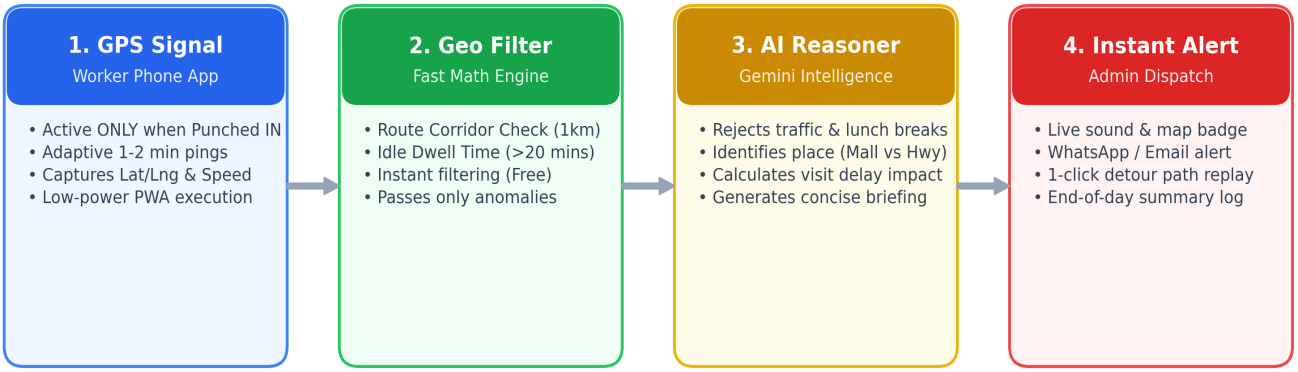
1. Visual Concept: How the AI Detects Detours & Unplanned Stops



2. Core Capabilities & What the System Tracks

FEATURE	HOW IT WORKS	CLIENT BUSINESS BENEFIT
Virtual Route Corridor	Creates a dynamic 500m - 1km safe travel corridor between assigned customer visits.	Allows minor lane changes or normal city road navigation without triggering false alarms.
Personal Stop & Dwell Detection	Detects when a worker remains stationary (> 20 mins) outside any scheduled customer premises.	Stops employees from wasting work hours at cafes, malls, or home during paid shifts.
Traffic vs. Detour Separation	AI cross-references road direction and speed: slow speed along route = traffic; divergent vector = detour.	Zero false alarms. Admins are only notified when real unauthorized deviations happen.
Instant Admin Dispatch	Real-time browser sound/toast alert + optional automated WhatsApp/Email notification.	Managers can take immediate action or call the worker before customer appointments are missed.

3. End-to-End Autonomous AI Workflow



4. Sample Incident Notification (What Company Admin Receives)

[!] REAL-TIME AI INCIDENT ALERT | Severity: HIGH | Timestamp: 11:18 AM

Worker: Suresh Kumar (Emp ID: FW-104) | **Branch:** Chennai Central Hub
Status: Unauthorized Detour & Dwell Detected
Details: Suresh deviated **3.4 km off-corridor** heading away from scheduled Client B (TechCorp Ltd). The device has been stationary at **'Phoenix Grand Mall, Velachery'** for **38 minutes**.
AI Schedule Impact: Scheduled arrival at TechCorp was 11:30 AM. Current delay estimate is **+45 mins**.
Recommended Action: Contact employee immediately or reassign urgent afternoon lead.
[Click to Open Live Replay Map] • [Log Formal Explanation Request]

5. Critical Technical Guarantees for Client Peace of Mind

Adaptive Battery Optimization	GPS sampling runs at 1-min intervals when traveling (>10 km/h) and shifts to sleep mode when stationary, consuming less than 4% phone battery over an 8-hour shift.
100% Employee Privacy Compliance	Tracking operates strictly between **Punch IN** and **Punch OUT**. Geolocation is completely cut off before punch in, during official breaks, and after punch out.
Lunch & Meal Break Safe Harbor	Pre-configured break windows (e.g., 1:00 PM - 2:00 PM) are recognized by the AI and logged as designated meal halts, preventing awkward false alarms.

6. Recommended 3-Phase Rollout Plan

PHASE	DELIVERABLES & SCOPE	ESTIMATED TIMELINE
Phase 1: Fast MVP	Route Corridor geometry checks, stationary idle flags (>20m), and live Red Warning markers on the Admin Map.	**3 to 5 Days**
Phase 2: AI Reasoning	Gemini AI integration for location classification, traffic vs detour discrimination, and concise incident briefs.	**4 to 6 Days**
Phase 3: Automation	Instant WhatsApp/Email notifications to Company Admins and End-of-Day Field Discipline Audit Reports.	**3 to 4 Days**